

MICROCONTROLLER-BASED MOVING-MESSAGE DISPLAY

■ PANKAJ KISHOR VERMA

LED-based moving-message displays are becoming popular for transmitting information to large groups of people quickly. These can be used indoors or outdoors. We can find such displays in areas like railway platforms, banks, public offices, hotels, training institutes, nightclubs and shops.

Compared to LEDs, liquid-crystal displays (LCDs) are easy to interface with a microcontroller for displaying information as these have many built-in functions. But these can't be observed from a distance and large-size LCDs are very costly.

LED-based displays can be of two types: dot-matrix and segmental. If you implement a moving-message display with multiplexed dot-matrix LEDs, it will be very costly for displaying 16 characters or more at a time. Moreover, programming will require a lot of data memory or program memory space. An external RAM may be needed to complement a microcontroller like AT89C51.

However, if you use alphanumeric (16-segment LED) displays for the above purpose, programming burden is reduced and also it becomes highly cost-effective. You can make your own display panel consisting of 16 alphanumeric characters at a much lower cost.

The circuit presented here uses 16 common-anode, single-digit, alphanumeric displays to show 16 characters at a time. Moreover, programming has been done to make the characters move in a beautiful manner. A message appears on the panel from the right side, stays for a few seconds when the first character reaches the leftmost place and then goes out from the left side.

It displays 16 different messages to depict different occasions, which can be selected by the user through a DIP switch.

Circuit description

Fig. 1 shows the circuit of the microcontroller-based moving-message display. It comprises microcontroller AT89C51, three-to-eight decoder 74LS138, common anode alphanumeric displays, regulator 7805 and a few discrete components.

At the heart of the moving-message display is Atmel AT89C51 microcontroller (IC1). It is a low-power, high-performance, 8-bit microcontroller with 4 kB of flash programmable and erasable read-only memory (PEROM) used as on-chip program memory, 128 bytes of RAM used as internal data memory, 32 individually



programmable input/output (I/O) lines divided into four 8-bit ports, two 16-bit programmable timers/counters, a five-vector two-level interrupt architecture, on-chip oscillator and clock circuitry.

Ports P0 and P2 of the microcontroller have been configured to act as a common data bus for all the 16 alphanumeric displays whose corresponding data pins have been tied together to make a common 16-bit data bus. Port-2 provides the higher byte of data, while port-0 provides the lower one to light up a character on the display. Port pins P1.2-P1.4 and P1.5-P1.7 of the microcontroller have been used as address inputs for decoder IC3 and IC4 (74LS138) to enable one of the fourteen alphanumeric displays (DIS3 through DIS16).

PARTS LIST

Semiconductors:

IC1	- AT89C51 microcontroller
IC2, IC3	- 74LS138 3-to-8 decoder
IC4	- 7805 5V regulator
T1-T16	- BC558 npn transistor
D1-D4	- 1N4007 rectifier diode
DIS1-DIS16	- KLA51 common-anode alphanumeric display
LED1	- 5mm LED

Resistors (all 1/4-watt, 5% carbon):

R1-R16	- 2.2-kilo-ohm
R17-R32	- 120-ohm
R33-R37	- 10-kilo-ohm
R38	- 220-ohm

Capacitors:

C1, C2	- 33pF ceramic disk
C3	- 2200µF, 25V electrolytic
C4	- 1µF, 16V electrolytic
C5	- 10µF, 16V electrolytic
C6	- 0.1µF ceramic disk

Miscellaneous:

X1	- 220V AC primary to 9V, 500mA secondary transformer
X _{TAL}	- 11.0592MHz crystal
S0-S3	- 4-pin DIP switch
S4	- Push-to-on switch

List of Messages Which can be Selected

S3S2S1S0	Message selected
0 0 0 0	Happy Birthday
0 0 0 1	Happy New Year
0 0 1 0	*Happy Diwali*
0 0 1 1	Merry Christmas
0 1 0 0	*Happy Holi*
0 1 0 1	*Eid Mubarak*
0 1 1 0	Happy Dashaheera
0 1 1 1	Happy Wedding
1 0 0 0	Happy Janmashtmi
1 0 0 1	*Happy Rakhi*
1 0 1 0	*Happy Pongal*
1 0 1 1	Happy Mothers Day
1 1 0 0	*Happy Ramjan*
1 1 0 1	*Happy Lohri*
1 1 1 0	*Happy Easter*
1 1 1 1	Welcome to All